RMO(INDIA) – 1991

Max Mark – 100 Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all the questions.

1. Let P be an interior point of a triangle ABC and AP, BP, CP meets the sides BC, CA and AB in point D, E and F respectively. Show that $\frac{AP}{PD}=\frac{AF}{FB}+\frac{AE}{EC}$.
2. If a, b, c and d are any four positive real numbers, then prove that $\frac{a}{b}+\frac{b}{c}+\frac{c}{d}+\frac{d}{a}\geq 4$.
3. A four digit number has the following properties:
   (a) It is a perfect square.
   (b) Its first two digits are equal to each other.
   (c) Its last two digits are equal to each other.
   Find all such four digit numbers.
4. There are two urns each containing an arbitrary number of balls. (Both are nonempty to begin with). We are allowed two types of operations:
   (a) Remove an equal number of balls in any one of them.
   (b) Double the number of balls in any of them.
   Show that after performing these operations finitely many times, both the urns can be made empty.
5. Take any point P_1 on side of BC of a triangle and draw following chain of lines; P_1P_2 parallel to AC, P_2P_3 parallel to BC; P_3P_4 parallel to AB; P_4P_5 parallel to CA; P_5P_6 parallel to BC. Here P_2, P_5 lie on AB; P_3, P_6 lie on CA and P_4 lie on BC. Show that P_6P_1 is parallel to AB.
6. Find all integers value of 'a' such that the quadratic expression $(x+a)(x+1991)+1$ can be factored as $(x+b)(x+c)$ where b and c are integers.
7. Prove that n^4+4^4 is composite for all integer values of n greater than 1.
8. The 64 square of 8X8 chess board are foiled with positive integers in such a way that each integer is the average on the neighborhood squares. Two square are neighbors, if they have a common edge or a common vertex. Thus, a square can have 8, 5, or 3 neighbors depending on its position. Show that all the 64 entries in fact equal.

******
****
**